

12.3 Latent Diffusion and Text Conditioning

Diffuse in a VAE latent. Condition on a text embedding (often CLIP from Week 5).

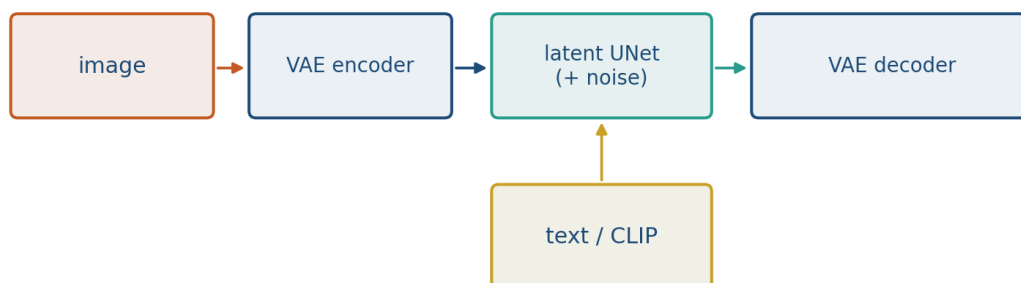
These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Pixel-space diffusion on 256×256 RGB is expensive. **Latent diffusion** (Rombach et al.) denoises in a compressed space. **Text** enters as a condition, usually from a frozen CLIP text tower (Week 5).

1. Diffuse in latent space

A pretrained encoder E maps an image to $z_0 = E(x)$. You run the forward process of note **12.1** on z , not on pixels. A decoder D_{VAE} turns the final latent back into an image. Training the U-Net on latents is cheaper; the VAE already threw away some high-frequency detail.

Latent diffusion: denoise a cheap latent, condition on text



Stable Diffusion is this picture plus a big text-conditioned U-Net. The idea is the same at homework scale: compress, denoise, decode.

Typical sizes to say out loud: an image $512 \times 512 \times 3$ becomes a latent like $64 \times 64 \times 4$. You pay diffusion steps on the small tensor.

2. Conditioning with CLIP text

You want $p(x \mid \text{caption})$, not an unconditional generator. Encode the caption with CLIP's **text** tower (note **5.1**). Inject that vector by:

- concatenating it to the U-Net's timestep embedding, or
- **cross-attention**: image (or latent) tokens query the text tokens.

Classifier-free guidance trains with and without the text, then at sample time does

$$\hat{\epsilon} = \epsilon_{\theta}(z_t, t, \emptyset) + s \left(\epsilon_{\theta}(z_t, t, c) - \epsilon_{\theta}(z_t, t, \emptyset) \right).$$

Scale $s > 1$ sharpens adherence to the prompt and can wash out diversity. Report s .

CLIP scores pairs; it does not decode a caption (note **5.2**). Here you only need the text embedding as a handle.

3. What to measure

Text-to-image is not “the picture looks nice.” Report a prompt-alignment score (CLIP cosine of image vs text), a coverage or diversity check, and a failure case (wrong count, wrong object, reading text in the image). Conditioning does not remove Week 5’s bias story: the same CLIP geometry is in the loop.

4. Teaching this note

About **35 minutes** at the board, then **~12 minutes** of selected video from the first 45 minutes (do not play the five-hour coding rest).

- **0–12 min.** Why latents: VAE encode / denoise / decode. Count pixels vs latent cells.
 - **12–22 min.** CLIP text tower as c . Cross-attention in one sentence.
 - **22–33 min.** Classifier-free guidance with numbers. Diversity vs prompt match.
 - **Then** play Umar Jamil Stable Diffusion **4:30–12:07** (what SD is; forward/reverse in this stack) and **22:20–31:00** (CFG). Optional 31:00–39:54 (CLIP + VAE + text-to-image) if time. Stop by **45:00**.
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5. Worked example

Latent size. A $512 \times 512 \times 3$ image is 786,432 numbers. A $64 \times 64 \times 4$ latent is 16,384 numbers: about $48 \times$ fewer. Fifty reverse steps on the latent are cheaper than fifty steps on pixels.

Guidance. Let the net’s two noise guesses (one scalar, to see the algebra) be $\varepsilon_{\theta}(z_t, t, \emptyset) = 0.8$ (unconditional) and $\varepsilon_{\theta}(z_t, t, c) = 0.2$ (with prompt c). The difference is $0.2 - 0.8 = -0.6$.

$$\hat{\varepsilon} = 0.8 + s(-0.6) = 0.8 - 0.6s.$$

s	$\hat{\varepsilon}$	reading
1	0.2	just the conditional net
3	−1.0	step farther along the prompt direction
7.5	−3.7	very sharp prompt match; variety usually drops

$s = 1$ is “use the text-conditioned prediction.” $s > 1$ **extrapolates** away from the unconditioned guess. That is why a huge s can look like a poster of the prompt and still miss modes (Week 11’s coverage lesson, now in prompt space).

6. Where students get stuck

- Fine-tuning CLIP's image tower at sample time. Text-to-image needs the **text** tower (and a VAE decoder). The image tower is for scoring pairs, not for sampling.
 - Calling $s = 1$ "no text." $s = 1$ is still conditional; the empty prompt $\emptyset$ is the other call.
 - Measuring only FID or only "looks nice," with no prompt-alignment number and no failure case.
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7. Video

Watch [Umar Jamil, Coding Stable Diffusion from scratch in PyTorch](#), **0:00–45:00** assigned, with in-class pauses at **4:30–12:07** (what Stable Diffusion is; forward/reverse) and **22:20–39:54** (classifier-free guidance, CLIP, VAE, text-to-image).

Do not start the coding chapters (VAE implementation from 44:30 onward) in this lecture. The board is the picture; the lab is still 2-D denoising from notes **12.1–12.2**.

8. Practice

1. Why denoise latents instead of pixels if you already have a VAE?
2. CLIP's image tower is optional at sample time for text-to-image. Which CLIP tower is not?
3. Guidance scale $s = 1$ versus $s = 12$: what do you expect to happen to prompt match and to variety?
4. $\epsilon_{\text{uncond}} = 1.0$, $\epsilon_{\text{cond}} = 0.4$, $s = 5$. Compute $\hat{\epsilon}$.
5. A $256 \times 256 \times 3$ image vs an $32 \times 32 \times 4$ latent: how many numbers in each, and what is the ratio (image / latent)?